Penguins

2025-09-26

Introduction

This is a simple analysis of the Penguins dataset from the Palmer Penguins dataset. This analysis will go over linear regression, data visualization, anova, and t-testing. If you would like at the code for this project, please visit the markdown file.

Libraries

Here are the libraries used for this viusualization and analysis.

```
library(datasets)
library(ggplot2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(tibble)
library(palmerpenguins)
library(BSDA)
```

```
## Loading required package: lattice
##
## Attaching package: 'BSDA'
##
## The following object is masked from 'package:datasets':
##
##     Orange
```

```
library(ggcorrplot)
```

```
## Warning: package 'ggcorrplot' was built under R version 4.4.3
```

Our Dataset

This dataset contains 344 observations with 7 factors:

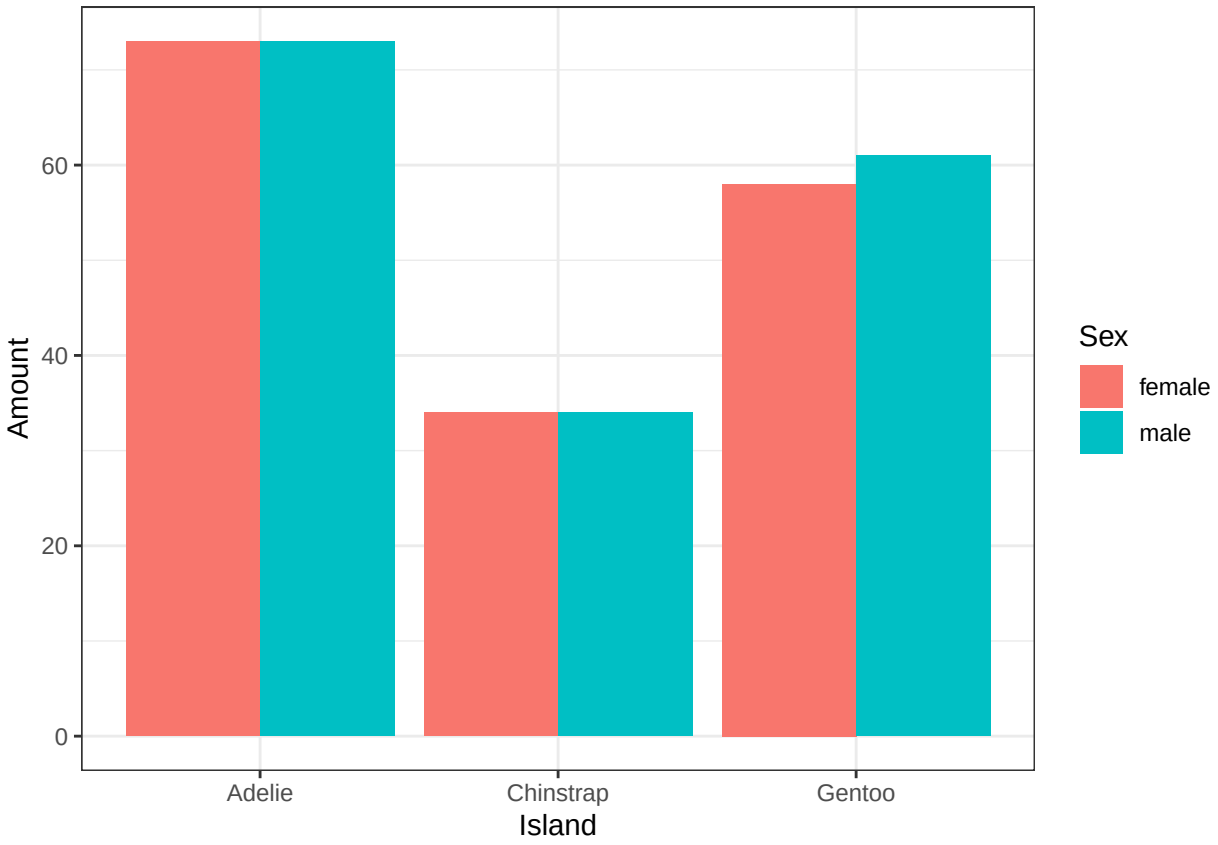
Species: Denotes penguin species
Island: Denotes an island in the Palmer Archipelago where the penguin was observed.
Bill_length_mm: Bill length in millimeters
Bill_depth_mm: Bill Depth in millimeters
Flipper_length_mm: Flipper Length in millimeters
Body_mass_g: Body mass in grams
Sex: Denotes penguin sex (Male/Female)
Year: Denotes study year(2007,2008, or 2009)

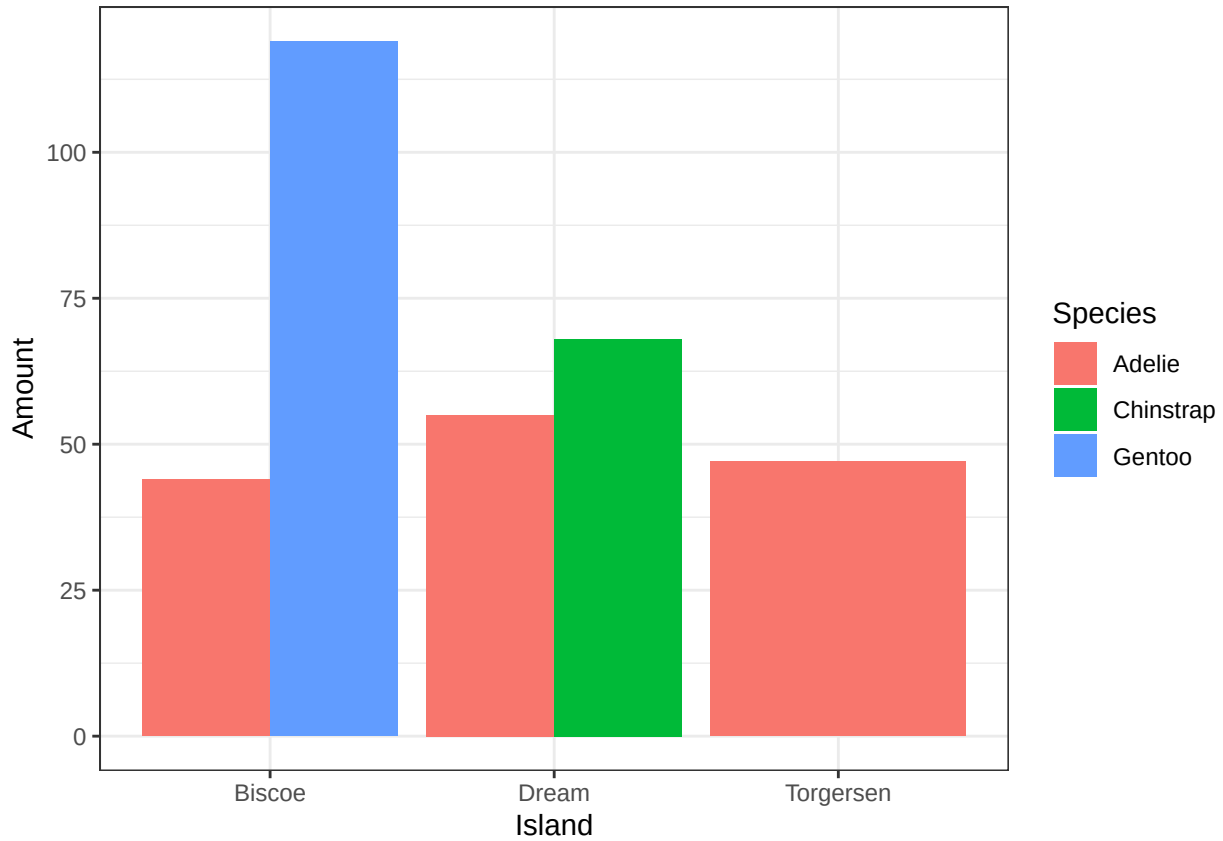
```
##       species          island    bill_length_mm  bill_depth_mm
##  Adelie   :152   Biscoe   :168   Min.   :32.10   Min.   :13.10
##  Chinstrap: 68   Dream    :124   1st Qu.:39.23   1st Qu.:15.60
##  Gentoo   :124   Torgersen: 52   Median :44.45   Median :17.30
##                                   Mean   :43.92   Mean   :17.15
##                                   3rd Qu.:48.50   3rd Qu.:18.70
##                                   Max.   :59.60   Max.   :21.50
##                                   NA's   :2       NA's   :2
##  flipper_length_mm  body_mass_g       sex           year
##  Min.   :172.0     Min.   :2700   female:165   Min.   :2007
##  1st Qu.:190.0     1st Qu.:3550   male  :168   1st Qu.:2007
##  Median :197.0     Median :4050   NA's  : 11   Median :2008
##  Mean   :200.9     Mean   :4202                Mean   :2008
##  3rd Qu.:213.0     3rd Qu.:4750                3rd Qu.:2009
##  Max.   :231.0     Max.   :6300                Max.   :2009
##  NA's   :2         NA's   :2
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

Initial Visualizations

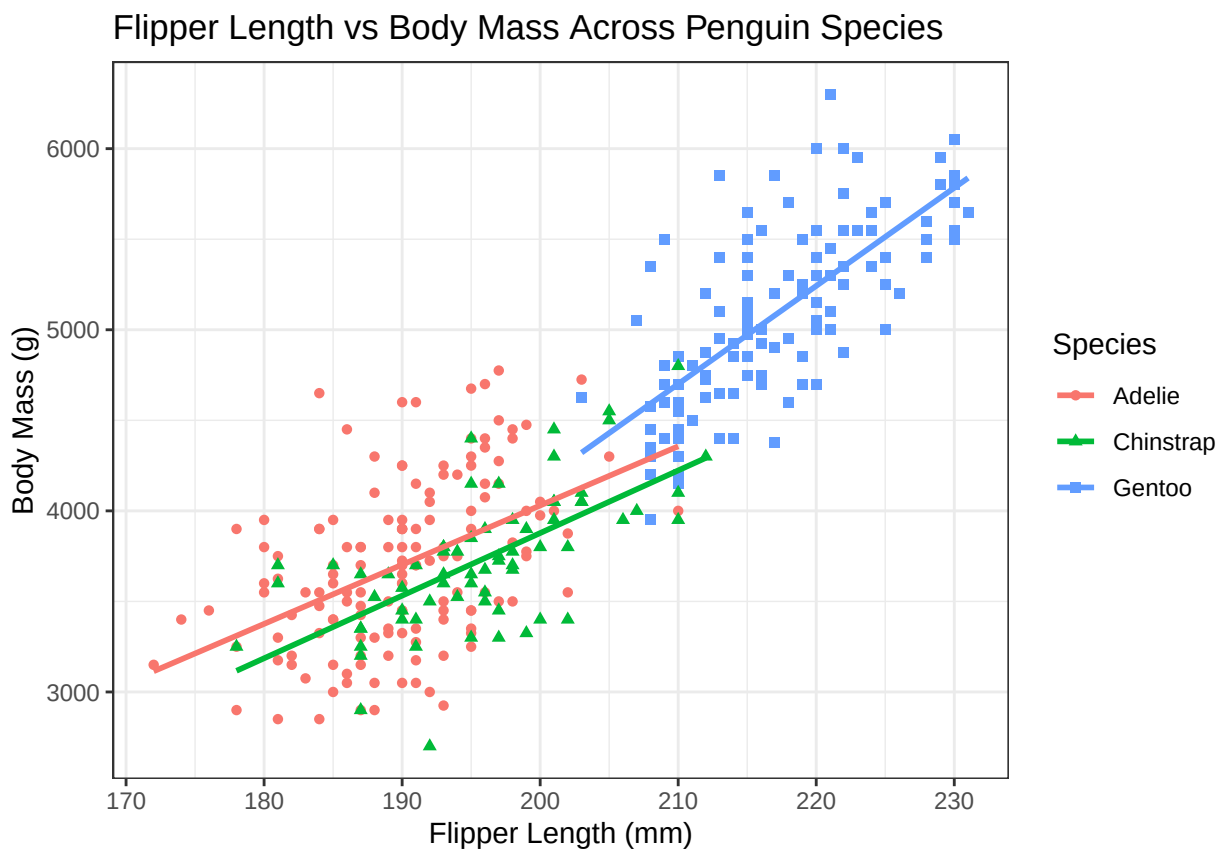
Before we dive into some regression analysis, let's get an overall understanding of what our data looks like and how it is distributed. With these graphs, we can see that we have a roughly equal amount of male and female penguins. Adelie penguins are the most plentiful, and are also the only species found on every island.


Amount
60
40
20
0
Adelie
Chinstrap
Gentoo
Island
Sex
female
male

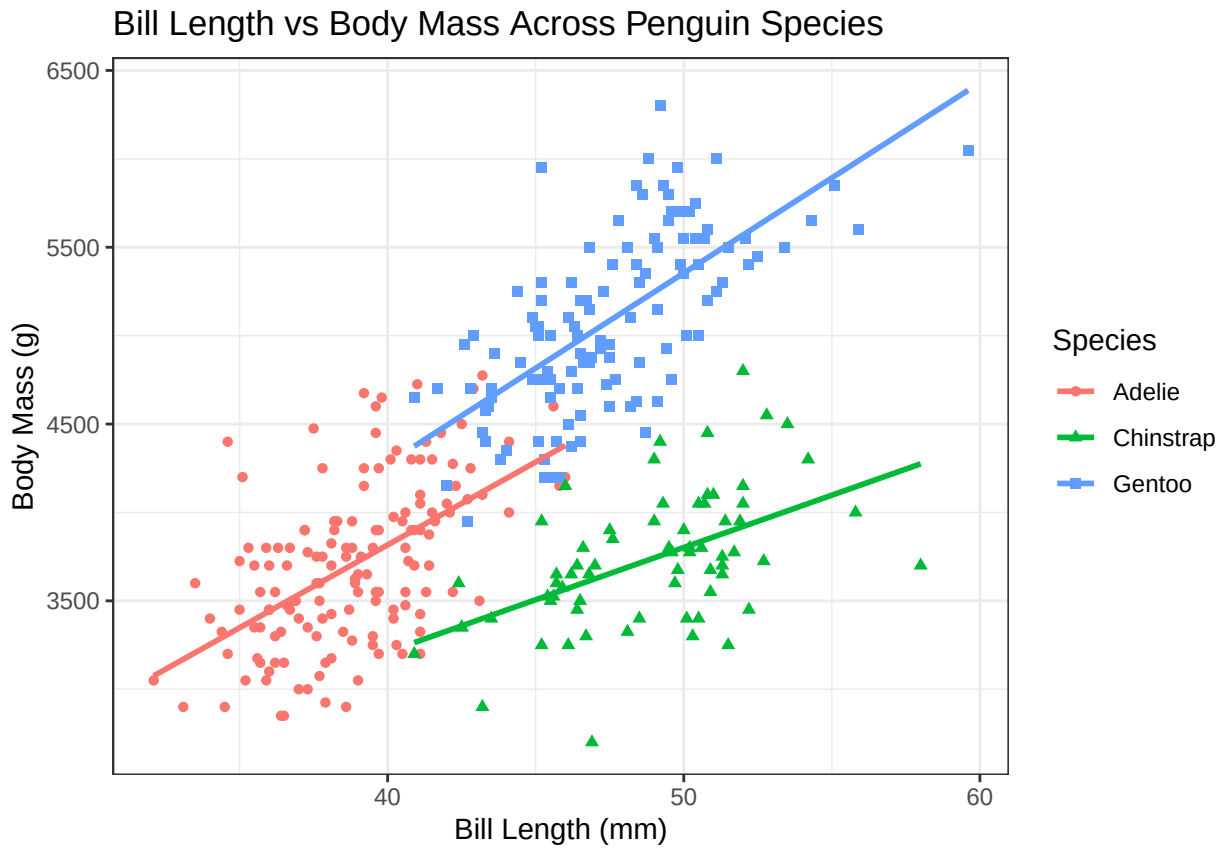


Suppose we want to analyze if there analyze the factors behind penguin mass. The following are graphs depicting the relationship between body dimensions and body mass across all species of penguins. Adelie and Chinstrap Penguins seem to have similar dimensions in regards to flipper length and bill depth. We can also see that there seems to be some positive relationship between body dimenstions and body mass.

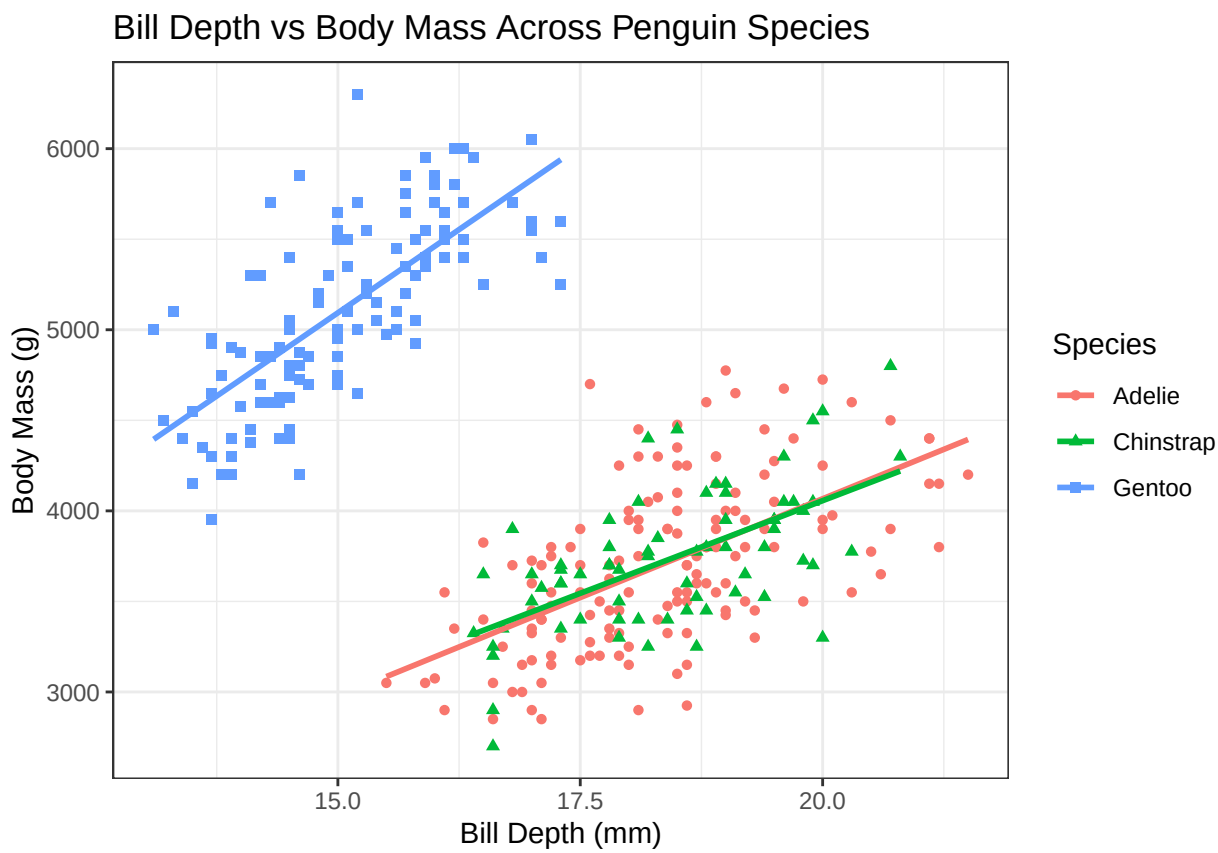
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

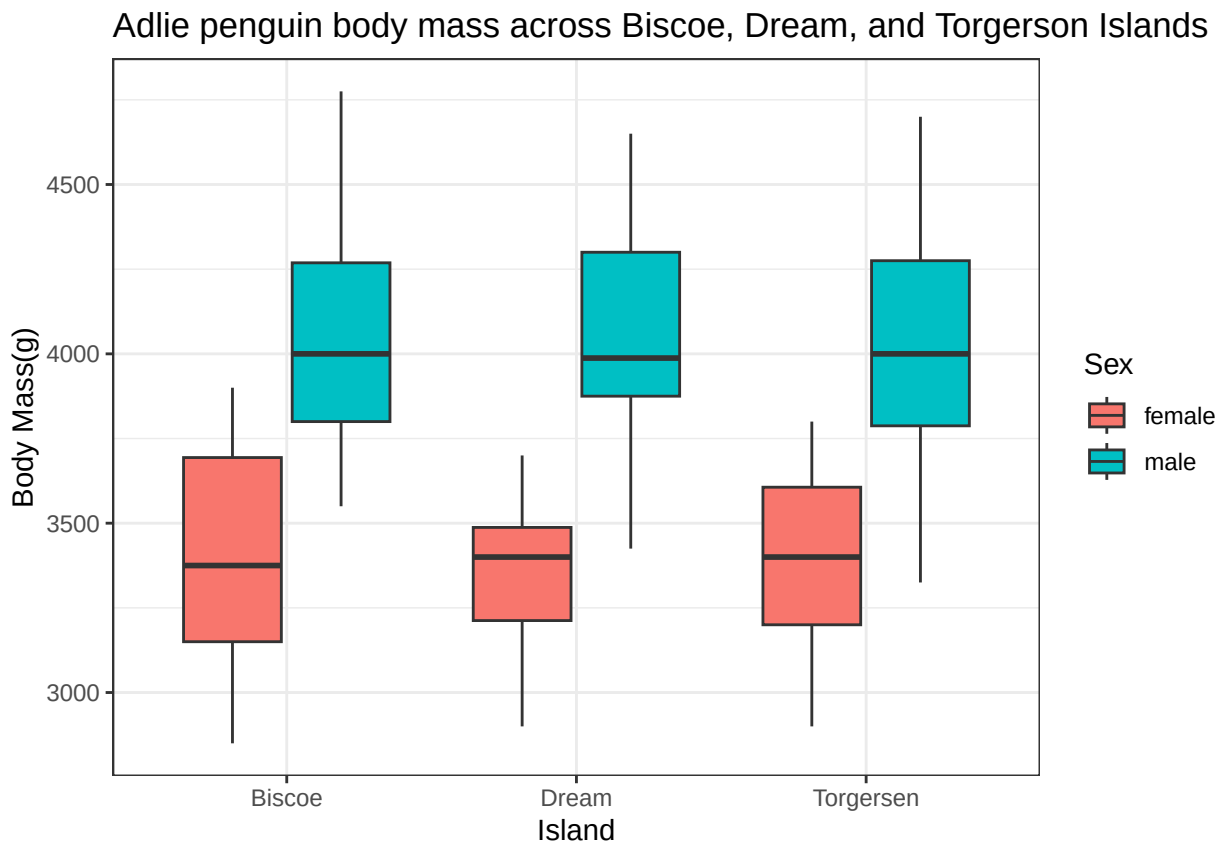


Since the Gentoo seem to be isolated from the other species regarding body mass, let's focus on them for now. LEt's start with a simple regression analysis using flipper length. If we do a simple regression analysis, we can see that flipper length is a significant predictor for body mass, with an extremely small p-value wellbelow our significance level of 0.05. We can reject the null and assume the alternative is true: flipper length is a significant predictor for body mass. With ever 1mm increase with flipper length for a Gentoo penguin, we can expect body mass to increase by 54 grams.

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = gentoo.penguins)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -704.69 -244.29  -58.87  161.98 1003.65
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -6674.204   1075.436  -6.206 8.51e-09 ***
## flipper_length_mm    54.165      4.948  10.946  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 354 on 117 degrees of freedom
## Multiple R-squared:  0.506,  Adjusted R-squared:  0.5017
## F-statistic: 119.8 on 1 and 117 DF,  p-value: < 2.2e-16
```

Suppose we wanted to compare the mass of the same species penguins across islands. Perhaps the

habitat of each island could have an effect on the penguins, causing them to be larger, for example. For this analysis, let's focus on only Adelie penguins, since they are the only species found across all three islands. From this boxplot, we can see that there initially seems to be quite a different in mass between male and female penguins, and only a small difference between islands.



Using ANOVA in R confirms these results: we result in a p-value well above our p-value of 0.05. We fail to reject the null and cannot confirm the alternative to be true. There is not enough evidence to assume the island has an effect on the Adelie penguin's mass.

```
##              Df   Sum Sq Mean Sq F value Pr(>F)
## island        2     2064    1032   0.005  0.995
## Residuals   143 30496138  213260
```

Now let's see if there is a significant difference in mass for the sex of an Adelie penguin. We can use a simple t-test for this. In these results, we can see that the mean mass in grams for males is significantly larger than females. We additionally have a p-value of nearly 0, meaning that we reject the null and confirm the alternative as true. There is a statistically significant difference between the mass of male and female adelie penuins.

```
##
##  Welch Two Sample t-test
##
## data:  body_mass_g by sex
## t = -13.126, df = 135.69, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group female and group male is not equal to
## 95 percent confidence interval:
##  -776.3012 -573.0139
```

```
## sample estimates:
## mean in group female   mean in group male
##             3368.836             4043.493
```